

Chapter 5: Differentiation

Section: The Continuity of Derivatives

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation

The Intermediate Value Property

Proof of Theorem 5.12

Consequences

Summary

Motivation



How Wild Can a Derivative Be?

- A function f can be differentiable at **every** point, yet f' can be **discontinuous** (Example 5.6(b))

How Wild Can a Derivative Be?

- A function f can be differentiable at **every** point, yet f' can be **discontinuous** (Example 5.6(b))
- Natural question: is *every* function a derivative? **No!**

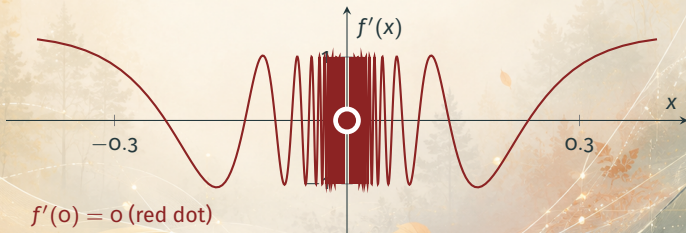
How Wild Can a Derivative Be?

- A function f can be differentiable at **every** point, yet f' can be **discontinuous** (Example 5.6(b))
- Natural question: is *every* function a derivative? **No!**
- Key fact: derivatives on an interval assume **intermediate values**, just like continuous functions (compare Theorem 4.23)

Recall Example 5.6(b): A Discontinuous Derivative

Example 5.6(b)

$f(x) = x^2 \sin(1/x)$ for $x \neq 0$, $f(0) = 0$: differentiable everywhere, but $f'(x) = 2x \sin(1/x) - \cos(1/x)$ has **no limit** as $x \rightarrow 0$.



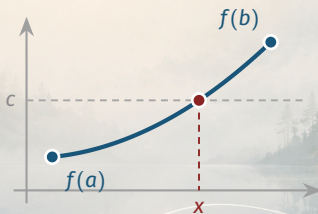


The Intermediate Value Property

Recall: The Intermediate Value Theorem

Recall: Theorem 4.23

If f is *continuous* on $[a, b]$ and $f(a) < c < f(b)$, then $f(x) = c$ for some $x \in (a, b)$.



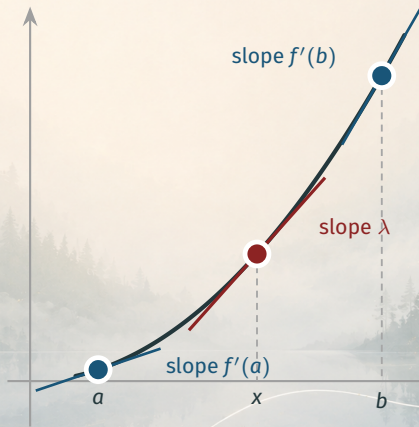
Theorem 5.12: Derivatives Assume Intermediate Values

Theorem 5.12 (Darboux)

Suppose f is a real differentiable function on $[a, b]$ and $f'(a) < \lambda < f'(b)$. Then there is a point $x \in (a, b)$ such that $f'(x) = \lambda$.

- A similar result holds if $f'(a) > f'(b)$
- **Note:** we assume **nothing** about the continuity of f' !

Theorem 5.12: The Picture





Proof of Theorem 5.12

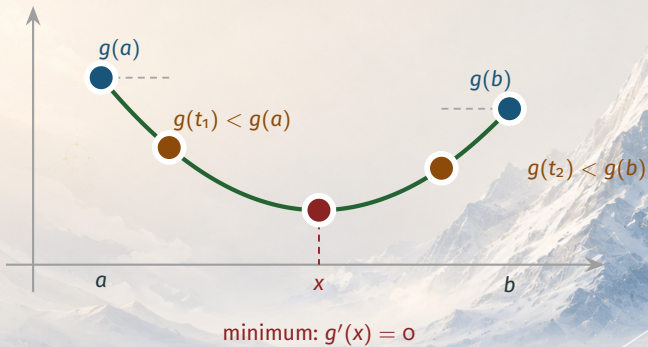
Proof Strategy: Theorem 5.12

Goal: Find $x \in (a, b)$ with $f'(x) = \lambda$.

Approach:

1. Tilt the picture:
study $g(t) = f(t) - \lambda t$, so the target slope becomes $g'(x) = 0$
2. Show neither endpoint can be a minimum of g (using $g'(a) < 0 < g'(b)$)
3. g is continuous on the compact set $[a, b]$, so it attains a minimum (Theorem 4.16) — necessarily at an **interior** point
4. At an interior minimum, $g'(x) = 0$ (Theorem 5.8)

Proof: The Picture



Proof of Theorem 5.12

Step 1: The tilted function

Put $g(t) = f(t) - \lambda t$. Then g is differentiable on $[a, b]$ and

$$g'(t) = f'(t) - \lambda, \quad g'(a) < 0, \quad g'(b) > 0.$$

Proof of Theorem 5.12

Step 2: a is not a minimum of g

$g'(a) < 0$ means $\lim_{t \rightarrow a} \frac{g(t) - g(a)}{t - a} < 0$, so for some $t_1 \in (a, b)$:

$$g(t_1) < g(a)$$

Proof of Theorem 5.12

Step 3: b is not a minimum of g

Similarly, $g'(b) > 0$ means $\lim_{t \rightarrow b} \frac{g(t) - g(b)}{t - b} > 0$, and now $t - b < 0$, so for some $t_2 \in (a, b)$:

$$g(t_2) < g(b)$$

Proof of Theorem 5.12

Step 4: g attains an interior minimum

g is continuous on the compact set $[a, b]$, so it attains its minimum at some point $x \in [a, b]$ (Theorem 4.16). By Steps 2 and 3, $x \neq a$ and $x \neq b$:

$$a < x < b$$

Proof of Theorem 5.12

Step 5: Conclusion

g has a local minimum at the interior point x and $g'(x)$ exists, so $g'(x) = 0$ (Theorem 5.8). Hence

$$f'(x) = \lambda$$



Consequences



Recall: Two Kinds of Discontinuity

Recall: Definition 4.26

Let f have a discontinuity at x .

- **Simple discontinuity (first kind):** $f(x+)$ and $f(x-)$ both exist
- **Discontinuity of the second kind:** $f(x+)$ or $f(x-)$ fails to exist

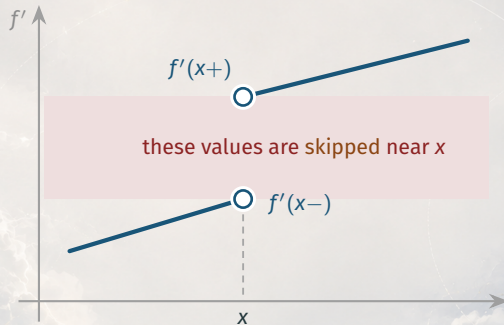
Corollary: No Simple Discontinuities

Corollary (to Theorem 5.12)

If f is differentiable on $[a, b]$, then f' cannot have any **simple** discontinuities on $[a, b]$.

Key idea: a jump in f' would force f' to **skip** intermediate values — impossible by Theorem 5.12.

Why a Jump in f' Is Impossible



Values between $f'(x-)$ and $f'(x+)$ are never attained near x — contradicting Theorem 5.12.

But the Second Kind Can Happen

Remark

f' may very well have discontinuities of the **second kind**.

- Example 5.6(b): $f'(x) = 2x \sin(1/x) - \cos(1/x)$ **oscillates** as $x \rightarrow 0$
- Neither $f'(0+)$ nor $f'(0-)$ exists — yet no intermediate value is skipped

The background is a soft-focus landscape featuring a calm lake reflecting distant mountains and cherry blossom trees. A complex, golden geometric pattern, consisting of interconnected lines and dots, is overlaid on the scene. A horizontal line is positioned below the word 'Summary'.

Summary

Summary

Key takeaways from this section:

- **Theorem 5.12 (Darboux):** if f is differentiable on $[a, b]$, then f' assumes every value between $f'(a)$ and $f'(b)$

Summary

Key takeaways from this section:

- **Theorem 5.12 (Darboux):** if f is differentiable on $[a, b]$, then f' assumes every value between $f'(a)$ and $f'(b)$
- **Corollary:** f' has **no simple discontinuities** — a derivative can never jump

Summary

Key takeaways from this section:

- **Theorem 5.12 (Darboux):** if f is differentiable on $[a, b]$, then f' assumes every value between $f'(a)$ and $f'(b)$
- **Corollary:** f' has **no simple discontinuities** — a derivative can never jump
- But f' can have discontinuities of the **second kind** (oscillation, as in Example 5.6(b))

Coming up next: L'Hospital's Rule — computing limits of quotients via derivatives.